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Sharif University
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Airport (20_582)
Homework 1

Problem 1

Problem Description

Consider a sequence of aircraft that are scheduled to land on a single runway. Let A_i denote the **planned landing time** of aircraft i.

Due to operational uncertainties, the **actual landing time** is subject to random error:

$$L_i = A_i + e_i$$

where the landing error is normally distributed:

$$e_i \sim \mathcal{N}(0, 6)$$

and errors are assumed to be independent across aircraft.

For safety reasons, the minimum required separation between two consecutive actual landings is:

$$L_{i+1} - L_i \geq 80$$

If this condition is violated, the aircraft must perform a go-around, reducing runway capacity.

The objective is to determine the optimal planned spacing between aircraft that maximizes expected runway capacity.

Distribution of Landing Separation

Define the actual landing time difference:

$$D_i = L_{i+1} - L_i$$

Substituting:

$$D_i = (A_{i+1} - A_i) + (e_{i+1} - e_i)$$

Assume constant planned spacing:

$$s = A_{i+1} - A_i$$

Since $e_i \sim \mathcal{N}(0, 6)$ and errors are independent:

$$e_{i+1} - e_i \sim \mathcal{N}(0, 12)$$

because variances add:

$$\mathrm{Var}(e_{i+1} - e_i) = 6 + 6 = 12$$

Therefore:

$$D_i \sim \mathcal{N}(s, 12)$$

Probability of Safe Landing

The probability that a landing is successful is:

$$P(D_i \geq 80)$$

Using standardization:

$$P(D_i \geq 80) = 1 - \Phi\left(\frac{80 - s}{\sqrt{12}}\right)$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function.

Capacity Function

Runway capacity is defined as the expected number of successful landings per unit time.

Since each landing consumes s units of time on average, the expected capacity is:

$$C(s) = \frac{1 - \Phi\left(\frac{80-s}{\sqrt{12}}\right)}{s}$$

This represents the expected number of successful landings per unit time.

Optimization Problem

The optimal planned separation s^* is obtained by solving:

$$\max_{s>0} \quad C(s) = \max_{s>0} \quad \frac{1 - \Phi\left(\frac{80-s}{\sqrt{12}}\right)}{s}$$

Since a closed-form solution does not exist, s^* must be computed numerically.

Final Capacity

After computing s^* numerically, the maximum runway capacity is:

$$C^* = C(87.471^*) = 0.011255$$

If time is measured in seconds, runway capacity in landings per hour is:

$$\text{Capacity per hour} = 3600 \times C^* = 40.518$$

Problem 2

Let:

- $T \sim \mathcal{N}(\mu, \sigma^2)$ be the total travel + airport processing time.
- H be the time between two consecutive flights (in minutes).
- x be the decision variable: time before the next flight at which the passenger leaves home.

If a flight departs at time 0, the passenger leaves at time $-x$. Arrival time at the gate is:

$$A = -x + T$$

Waiting Time Function

Two cases arise:

$$W(x, T) = \begin{cases} x - T & \text{if } T \leq x \\ H + x - T & \text{if } T > x \end{cases}$$

Equivalently,

$$W(x, T) = x - T + H\mathbf{1}_{\{T > x\}}$$

Expected Waiting Time

Taking expectation:

$$\mathbb{E}[W(x)] = x - \mu + H\mathbb{P}(T > x)$$

Since T is normally distributed,

$$\mathbb{P}(T > x) = 1 - \Phi\left(\frac{x - \mu}{\sigma}\right)$$

Therefore, the objective function is:

$$f(x) = x - \mu + H\left[1 - \Phi\left(\frac{x - \mu}{\sigma}\right)\right]$$

Optimization Problem

$$\min_{0 \leq x \leq H} f(x)$$

First Order Condition

Taking derivative:

$$f'(x) = 1 - \frac{H}{\sigma}\phi\left(\frac{x - \mu}{\sigma}\right)$$

Setting $f'(x) = 0$:

$$1 = \frac{H}{\sigma}\phi\left(\frac{x - \mu}{\sigma}\right)$$

Thus, the optimal x^* satisfies:

$$\phi\left(\frac{x^* - \mu}{\sigma}\right) = \frac{\sigma}{H}$$

This implicitly determines the optimal departure time.

Answers to Questions

(a) Optimal x when $H = 3$ hours

Convert hours to minutes:

$$H = 3 * 60 = 180 \text{ minutes}$$

Solve:

$$\phi\left(\frac{x^* - 72}{\sqrt{17}}\right) = \frac{\sqrt{17}}{180}$$

$$x_1^* = 65 \text{ minutes}$$

$$x_2^* = 79 \text{ minutes}$$

Since x must be greater than the 72(mean of wait time), the optimal departure time is:

$$x^* = 79 \text{ minutes}$$

The mean waiting time is: 15.06 minutes.

(b) Effect of Increasing H to 6 Hours

If H increases:

$$\phi(z^*) = \frac{\sigma}{H}$$

Since H increases, the right-hand side decreases. Because $\phi(z)$ decreases as $|z|$ increases from 0,

$$|z^*| \text{ increases}$$

Therefore,

$$x^* \text{ increases}$$

$$x_1^* = 63.481 \text{ minutes}$$

$$x_2^* = 80.518 \text{ minutes}$$

Conclusion: When flights are less frequent, the passenger departs earlier (larger safety buffer) -> 80.518.

The mean waiting time is: 15.51 minutes.

(c) Mean Number of People Waiting in Airport

Assume:

- D destinations
- Each destination has flight interval H
- Demand per destination = λ persons per flight

Total arrival rate per destination:

$$\lambda_d = \frac{\lambda}{H}$$

Using Little's Law:

$$L = \lambda_d \cdot \mathbb{E}[W(x^*)]$$

Total number waiting across all destinations:

$$L_{total} = D \cdot \lambda_d \cdot \mathbb{E}[W(x^*)]$$

$$L_{total} = 30 \cdot \frac{60}{180} \cdot \mathbb{E}[W(79)] = 150.60$$

Problem 3

Current situation

$$t = \frac{1}{40 - q}$$

t is mean time for service in Operation each hour.

$$total = \frac{q}{40 - q}$$

total is the service time, in one hour.

$$day = \frac{24 \times q}{40 - q}$$

all the service time, each day

$$year = \frac{365 \times 24 \times q}{40 - q}$$

all the service time, each year

$$cost = \frac{3000 \times 365 \times 24 \times q}{40 - q} = 26280000 \times \frac{q}{40 - q}$$

This is the yearly cost for current situation.

Future situation

$$t = \frac{1}{40 - (q/2)}$$

t is mean time for service in Operation each hour.

$$total = \frac{q}{40 - (q/2)}$$

total is the service time, in one hour for one Band.

$$day = \frac{24 \times q}{40 - (q/2)}$$

all the service time, each day for one Band.

$$year = \frac{365 \times 24 \times q}{40 - (q/2)}$$

all the service time, each year for one Band.

$$year = \frac{2 \times 365 \times 24 \times q}{40 - (q/2)}$$

all the service time, each year for both Band.

$$cost = \frac{3000 \times 2 \times 365 \times 24 \times q}{40 - (q/2)} = 52560000 \times \frac{q}{40 - (q/2)}$$

This is the yearly cost for future situation.

$$build = \frac{i \times (i+1)^{40}}{(i+1)^{40} - 1} \times 60M + 3M = 8031609.69(8.0316M)$$

This is the yearly cost build the new Band.

$$build = 8031609.69 + 52560000 \times \frac{q}{40 - (q/2)}$$

The total cost for future in reality.

Compare to find the best year

$$q = 20 \times (1 + 0.1)^n$$

Therefore current yearly cost in year n is:

$$current = 26280000 \times \frac{20 \times (1 + 0.1)^n}{40 - (20 \times (1 + 0.1)^n)}$$

Therefore future yearly cost in year n is:

$$future = 8031609.69 + 52560000 \times \frac{20 \times (1 + 0.1)^n}{40 - \frac{(20 \times (1+0.1)^n)}{2}}$$

We solve it for n=0:

$$26.28M \leq 43.07M$$

We solve it for n=1:

$$32.12M \leq 47.90M$$

We solve it for n=2:

$$40.25M \leq 53.62M$$

We solve it for n=3:

$$52.29M \leq 60.45M$$

We solve it for n=4:

$$71.80M \leq 68.72M$$

Therefore, before the start of **year 4**, we should to build the new Band.

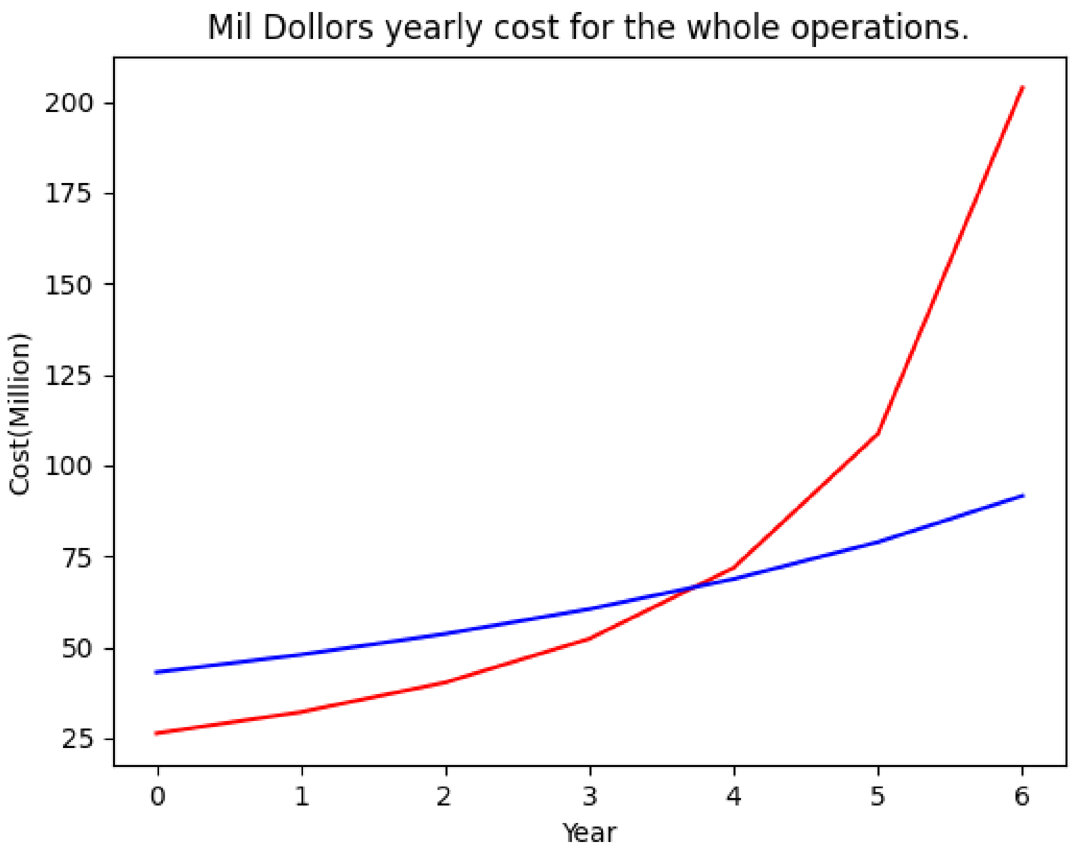

Figure 1: During 7 years, coparison the costs.